

LAKSHAY HASIJA

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Data Analyst with 2.5+ years of experience in the US energy sector, building data pipelines, automation systems, and ML solutions for natural gas pipeline operations. Proficient in Python, SQL, and Power BI, with hands-on experience in LLM tooling, drift monitoring, and statistical analysis. Adept at translating complex operational data into business insights and leading cross-functional analytics initiatives.

SKILLS

Languages & Tools	: Python (Pandas, NumPy, Selenium), SQL, LangChain, FastAPI, Streamlit, Azure, Jenkins, Git
ML & AI	: Random Forest, Logistic Regression, NLP, Drift Detection, MLflow, GenAI, Feature Engineering, Time Series
Statistics & Experimentation	: Hypothesis Testing, A/B Testing, Confidence Intervals, Probability Distributions, Power Analysis
Databases	: MySQL, MS SQL Server
BI & Visualization	: Power BI, Tableau, Matplotlib, Seaborn

WORK EXPERIENCE

Midtown Software | US Natural Gas Pipeline & EDI Scheduling Platform | Mohali September, 2023 – Present

Data Analyst

- Team Leadership:** Manage and mentor a team of 4, guiding cross-functional collaboration to design data-driven decision frameworks and automate reporting workflows.
- Machine Learning & NLP:** Automated notice classification pipelines using Python and NLP models to extract insights from unstructured text data, achieving 90% automation coverage.
- Business Intelligence:** Engineered interactive Power BI dashboards and improved Excel-based reporting frameworks to track business KPIs, reducing manual reporting efforts by 40% and improving access to operational metrics across teams.
- Database Optimization:** Designed and optimized complex SQL ETL workflows across multiple data sources, reducing query complexity and improving report turnaround for pipeline operations data including tariffs, nominations, and capacity bids.
- Stakeholder Management:** Gathered requirements from leadership and delivered actionable insights through ad hoc analysis, data cleaning, and validation.

RELATED PROJECTS

Azure CI/CD & RPA Pipeline Automation (2026)

- Built a Python-based RPA bot to extract unsubscribed (UNSC) capacity data from multiple US natural gas pipeline websites and upload it to production via REST API and Cosmos DB.
- Implemented Jenkins CI/CD pipeline to automate scheduled execution, with intermediate storage checkpointing to prevent re-scraping on network failures.
- Optimized data processing logic reducing end-to-end pipeline runtime from around 3 hours to around 1 hour (70% performance improvement).

MS SQL Chatbot Integration (2024)

- Developed a Python-based chatbot integrated with MS SQL Server to serve real-time pipeline data queries on the company website for both internal teams and external customers.
- Enabled natural language lookups for operational data such as top bidders, nomination activity, and capacity availability, eliminating manual database queries.
- Reduced manual data lookup time by 60%, improving response speed for day-to-day scheduling and trading operations.

A/B Testing Framework (2025)

- Built a complete statistical experimentation framework from scratch supporting both Frequentist (Z-tests) and Bayesian (Beta-Binomial) approaches for flexible experiment design.
- Integrated Power Analysis to calculate minimum detectable effect sizes and required sample sizes before test launch, preventing underpowered experiments.
- Enabled the team to statistically validate product and feature changes with rigorous, reproducible methodology, replacing ad hoc analysis.

LLM-Powered SQL Agent (2026)

- Built a natural language-to-SQL assistant using LangChain and Gemini, allowing non-technical users to query databases in plain English without writing SQL.
- Surfaced query results through a Streamlit dashboard with AI-generated summaries, making data accessible across teams without analyst intervention.

ML Pipeline with Drift Monitoring (2025)

- Deployed a Random Forest churn prediction model (ROC-AUC: 0.785) as a production REST API using FastAPI, with MLflow for full experiment tracking and model versioning.
- Built a KS-test-based drift detection module that continuously monitors input feature distributions and triggers automated retraining alerts when significant drift is detected.

EDUCATION

- Ramanujan College, Delhi University** | B.Voc. Software Development | 2023 | 8.5CGPA
- Cambridge School, Srinivaspuri** | Class XII, Commerce with Maths | 2020 | 85% (CBSE)

CERTIFICATES

Python and SQL for Data Science (Scaler)
Power BI - Complete Microsoft Power BI - Advance (Udemy)